Association for Information Systems
AIS Electronic Library (AISeL)

AMCIS 2024 Proceedings

Americas Conference on Information Systems (AMCIS)

August 2024

Digital Nudging to Reduce Online Returns: Can a Simple Picture Save the Environment?

Caterina Rauh
University of Bamberg, caterina.rauh@uni-bamberg.de

Christian Straubert
University of Bamberg, christian.straubert@uni-bamberg.de

Eric Sucky
University of Bamberg, eric.sucky@uni-bamberg.de

Follow this and additional works at: https://aisel.aisnet.org/amcis2024

Recommended Citation

Rauh, Caterina; Straubert, Christian; and Sucky, Eric, "Digital Nudging to Reduce Online Returns: Can a Simple Picture Save the Environment?" (2024). *AMCIS 2024 Proceedings*. 6.
https://aisel.aisnet.org/amcis2024/sig_green/sig_green/6

This material is brought to you by the Americas Conference on Information Systems (AMCIS) at AIS Electronic Library (AISeL). It has been accepted for inclusion in AMCIS 2024 Proceedings by an authorized administrator of AIS Electronic Library (AISeL). For more information, please contact elibrary@aisnet.org.

Digital Nudging to Reduce Online Returns: Can a Simple Picture Save the Environment?

Completed Research Full Paper

Caterina Rauh
University of Bamberg
caterina.rauh@uni-bamberg.de

Christian Straubert
University of Bamberg
christian.straubert@uni-bamberg.de

Eric Sucky
University of Bamberg
eric.sucky@uni-bamberg.de

Abstract

The growth of e-commerce brings with it a new environmental challenge: product returns. Based on the self-determination theory (SDT), we investigate whether digital nudging is a useful tool to lower return rates. We study how nudging affects consumer's autonomy (A), competence (C), and relatedness (R) need satisfaction and how consumers' environmental consciousness (EC) influences the effects of nudging. A survey-based online experiment with online shoppers from the US (n=988) is analyzed using structural equation modeling (SEM). We show, for instance, that nudging has a strong direct effect on return motivation (RM), suggesting that it serves as an extrinsic motivator. However, we also found that nudging has unintended side effects on intrinsic motivation, resulting in lower purchase motivation (PM). Our results show that nudging can be a promising tool to avoid product returns, but that it also bears risks.

Keywords

Digital Nudging, Preventive Return Management, E-Commerce, Sustainability, Consumer Behavior.

Introduction

Global e-commerce sales continue to grow. In the US, online retail sales more than doubled between 2020 and 2023; rising from USD 565 billion to more than USD 1.4 trillion. Unfortunately, the growth of e-commerce has a dark side: product returns. In the USA alone, the total amount of returns from online sales in 2023 was USD 247 billion (ApprissRetail 2024). For online retailers (known as e-tailers), product returns result in lost sales and increased supply chain costs. However, the massive negative environmental impact is also noteworthy (Chaleshtari et al. 2022). Product returns increase CO_2 emissions and produce additional waste (Tian and Sarkis 2022). In the US, an estimated 24 million tons of CO_2 were generated by product returns in 2022. Moreover, returned goods are estimated to generate 9.5 billion pounds of landfill waste annually (Optoro 2022). It is clear that every returned product inevitably causes negative impacts on the environment.

A restrictive return policy (e.g., return fees or short return windows) is probably the most straightforward way to reduce the return rate. However, consumers may resent a tight return policy. Research shows that a lenient return policy (1) lowers the perceived purchase risk, (2) leads to a positive image and higher trust in the e-tailer, (3) thus increases sales, and (4) promotes customer loyalty (Rokonuzzaman et al. 2021). As a result, many e-tailers accept high return rates as part of their consumer-friendly, lenient return policies (Wang et al. 2019). For a sustainable solution of this trade-off, returns need to be managed in a way that reduces environmental impact without harming sales and customer loyalty. An important approach that helps to achieve both goals is to avoid returns before they occur. In this study, we analyze whether digital nudging can motivate consumers toward an eco-friendly return behavior.

Nudge theory – founded by Thaler and Sunstein (2008) – is grounded in the findings of psychological and behavioral economics research, which has shown that in many situations people do not behave according to rational economic theories. A nudge aims to steer individuals' behavior in a certain predictable direction, using gentle reinforcement and implicit suggestion to persuade them without restricting their choices (Thaler and Sunstein 2008). In e-commerce, digital nudging is used in particular as to motivate the purchase of (certain) products, to influence consumer behavior in a desired direction or to support customers in their purchasing decisions. The use of nudging in e-commerce as a means to promote sustainable consumer behavior in general has received considerable attention (Bergram et al. 2022). However, the potential use of digital nudging to decrease return motivation in particular has received limited research attention so far. Therefore, our main research question (RQ) is:

RQ: Can digital nudging motivate eco-friendly return behavior and thus reduce returns?

Our analysis builds on the psychology of human motivation, in particular the well-established self-determination theory (SDT) of Deci and Ryan (1985). The central idea of SDT is to focus on the type rather than the amount of motivation, which encompasses a continuum from intrinsic motivation to extrinsic motivation (Ryan and Deci 2000). Therefore, our analysis focuses not only on whether nudging influences return motivation in terms of sustainability, but also on how nudging influences return motivation. Based on the SDT, we are able to identify and evaluate spillover effects (e.g., positive and negative side effects) of nudging. In the following, we discuss nudging within the context of return management and develop our hypotheses (H) based on the SDT. We then present a survey-based online experiment with online shoppers from the US (n=988) and the results of a structural equation model (SEM). We conclude with a discussion, limitations, and an outlook.

Return Management and Digital Nudging in E-Commerce

Many reasons for product returns are logistical in nature (e.g., wrong delivery, delivery of damaged products, or late delivery) or related to consumer satisfaction. After receiving a product, customers compare their expectations, e.g., regarding appearance and quality, with the received product and are likely to return the product if their expectations are not met (Martínez-López et al. 2022). Typical reasons for returns related to customer satisfaction are (1) the product does not have the expected characteristics (e.g., wrong size or a different color than shown online), (2) the customer does not like the product (anymore), or (3) it was an impulse purchase and the customer now regrets the purchase (Ambilkar et al. 2022). At the same time, however, it is also true that a significant proportion of returns in online retail, especially for fashion products, are due to behavioral reasons: fraud and abuse of the return policy, such as bracketing and wardrobing. Preventive return management includes all measures and tools to prevent (gatekeeping) or avoid (avoidance) returns before the purchase, during the purchase process, and after the purchase (Brand and Kopplin 2023). Many measures aim to avoid returns caused by customer dissatisfaction. Examples include zoomable product images, videos, reviews, size guides (e.g., the size a model is wearing), or 3D scanning (e.g., for shoes). In this paper, we analyze whether an easy-to-implement instrument such as digital nudging can influence online shoppers' return motivation. Our proposed approach primarily aims to reduce behavioral and satisfaction-related returns (e.g., selection orders, impulse buying, or abusing lenient return policies) by nudging customers toward the environmental impact of high product returns.

Digital nudging is becoming increasingly important due to digitization and the ever-expanding range of decisions that individuals face online. Digital nudging describes the use of design elements to influence individuals' behavior in digital decision-making environments (Weinmann et al. 2016). E-commerce is one of the most important areas of application for digital nudging. The forms of digital nudging used are diverse and constantly evolving (Bergram et al. 2022). When nudges are designed to support environmentally friendly behavior, they are also referred to as green nudges. In recent years, interest in the use of green nudges has increased, particularly in the context of online shopping (Carrel et al. 2023). For example, Cossatin et al. (2024), Peters and Wieland (2023), and Schürmann et al. (2023) show that digital nudges can promote sustainable consumer behavior in online fashion retailing. Demarque et al. (2015) also find that nudging can lead to more sustainability in online shopping in general. However, it is worth noting that the online experiment by Mirbabaie et al. (2023) showed no significant effect of nudging on sustainable consumer behavior. One explanation for this result could be a perceived restriction of customers' freedom of choice due to nudging – that is, the treatment was offset by negative

side effects (Osman et al. 2020). Finally, Michels et al. (2022) show the potential of nudging to promote eco-friendly shipping options in an online shop. They conclude that nudges are effective in changing decisions, but may lead to a negative perception of the online shop by consumers. Two studies analyze the effects of nudging on product returns. Using a field experiment, the study by Ghose et al. (2023) compares the influence of different nudges on consumers' purchase and return behavior. The study shows that nudges lead to higher sales, but may be accompanied by higher product returns (negative side effect). Also in a field experiment, von Zahn et al. (2022) tested the effect of a green nudge that informs about the environmental problem associated with product returns. The study results show that using a simple, low-cost green nudge can significantly reduce product returns. However, the authors do not investigate possible (spillover) effects of green nudging on (future) purchases. As mentioned above, the use of nudges can have side effects that go beyond the originally intended effect. These spillover effects can influence behaviors that are not the target of the nudge (Nafziger 2020). Clot et al. (2022) emphasize that the type of motivation must be taken into account when analyzing and evaluating nudges. Our study aims to address this research gap by focusing on the type of motivation for product returns.

Theoretical Foundation and Research Model

Previous research shows a positive relationship between environmental consciousness and motivation for eco-friendly behavior (e.g., Khan 2023; Wang et al. 2021). However, motives like convenience and comfort disrupt or even change behavior, leading to a motivation-behavior gap (Geng et al. 2017). To address this problem, digital nudges are a promising method to increase consumers' motivation to behave sustainably and reconsider their return behavior (Demarque et al. 2015). In addition, understanding why individuals behave in certain ways is critical to effectively influence behavior. Deci and Ryan's (1985) SDT helps to understand what drives motivation by explaining the types of motivation ranging from intrinsic to extrinsic. According to SDT, individuals are intrinsically motivated to perform a certain behavior when the three basic psychological needs are fulfilled: (1) the need for autonomy (A), e.g., perceived decision freedom, (2) the need for competence (C), e.g., feedback on behavior, and (3) the need for relatedness (R), e.g., a sense of belonging (Ryan and Deci 2000). In the context of online shopping, the more these three basic psychological needs are satisfied, the higher the purchase and return motivation (PM/RM) because these actions are perceived as more enjoyable (***H1a/H2a***, see Figure 1 for our model hypotheses). Perceived decision freedom and a sense of competence should make returning items more enjoyable for customers, especially if this behavior is socially desirable. Similarly, perceived shopping autonomy and familiarity with the online shop should increase purchase motivation. At the same time, individual and contextual factors influence the effect of motivational stimuli. As the nudge changes the context of online shopping, the complexity of motivation could lead to unintended side effects. In particular, feelings of autonomy and competence may be hindered by contextual conditions, such as a nudge toward socially desirable behavior (Ryan and Deci 2000). If consumers are informed about which behavior is desirable, they will reconsider their behavior and try to conform to this desired behavior in order to avoid negative social consequences (Lin et al. 2013). Thereby, their perceived autonomy is reduced (Wachner et al. 2020). Consumers may also feel less competent and not develop a sense of belonging because their behavior is perceived as less desirable. Therefore, nudging could lead to a reduction in perceived autonomy, competence, and relatedness (***Hb***). This hypothesized reduction in intrinsic motivation leads us to the mediation hypotheses:

H1: *Nudging decreases autonomy, competence and relatedness need satisfaction (Hb) and thereby decreases return motivation (Hb mediated through H1a).*

H2: *Nudging decreases autonomy, competence and relatedness need satisfaction (Hb) and thereby decreases purchase motivation (Hb mediated through H2a).*

In simple terms, the green nudge might spoil the fun. That is, while customers may have less return motivation because of a guilty conscience, they may also feel less purchase motivation in a less enjoyable environment. However, we may have to reject these hypotheses if the nudge solely acts as an extrinsic motivator and does not decrease intrinsic motivation. Individuals try to promote a positive self-image by presenting themselves as "pro-environmental consumers" (Schubert 2017). Informing consumers about the negative consequences of returns can be seen as psychological pressure, as it highlights the consequences of their morally questionable behavior. Individuals want to act according to external moral and social values and may refrain from purchasing and returning at all (Bear et al. 2017). Thereby,

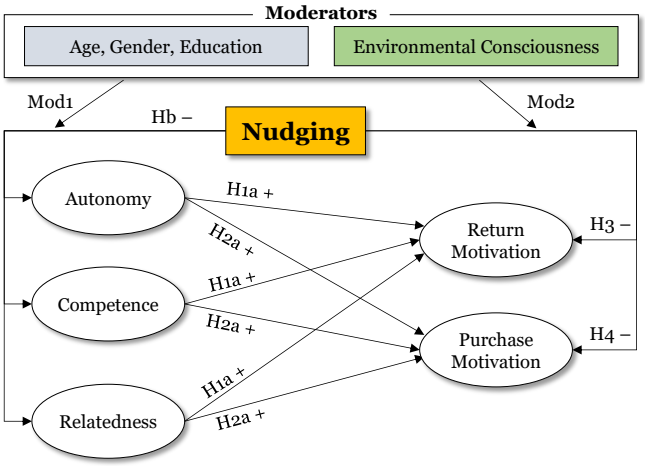


Figure 1. Model Hypotheses

nudging can also act as an extrinsic motivator that directly reduces RM and PM. Accordingly, we hypothesize that:

H3: *Nudging directly decreases return motivation.*

H4: *Nudging directly decreases purchase motivation.*

Motivation is promoted when a person supports or identifies with the value or importance of a behavior (Ryan et al. 2008). A nudge can appeal to motivation if it aligns with a person's values and interests. Thus, if a person is environmentally conscious, they may feel less offended by the green nudge because the nudge is consistent with their preferences. If someone shares the same values as the nudge reflects, then that person may feel less restricted in their autonomy and competence and may relate better to the online shop. Therefore, we expect the following moderation effects (Mod):

Mod1: *The more environmentally conscious a person is, the less green nudging decreases autonomy, competence, and relatedness need satisfaction.*

Mod2: *The more environmentally conscious a person is, the more [less] green nudging reduces their return motivation [purchase motivation].*

Research Method

Research Design, Data Collection, and Sample Characteristics

To evaluate the research questions and hypotheses, an online experiment was conducted with US citizens through "Prolific", a research platform for recruiting participants for online studies. Researchers upload their study to the Prolific website, estimate the time required, and set a reward for their project. Potential participants can then take part in the survey and get compensated for their participation. In our study, respondents were assigned to one of two experimental groups and then asked to imagine a fictitious online fashion shop called "Fashionstore24", which they were already familiar with. The participants were then introduced to a loyalty program called "Member Club". Similar to existing membership programs (e.g., Nike and H&M), the Fashionstore24 Member Club offers a lenient return policy (free shipping and a 100-day free return policy). While the control group (CG, n=493) was only presented with the general terms and conditions, the nudging group (NG, n=495) was exposed to additional information (i.e., the green nudge) about the negative impacts of product returns (see Figure 2).

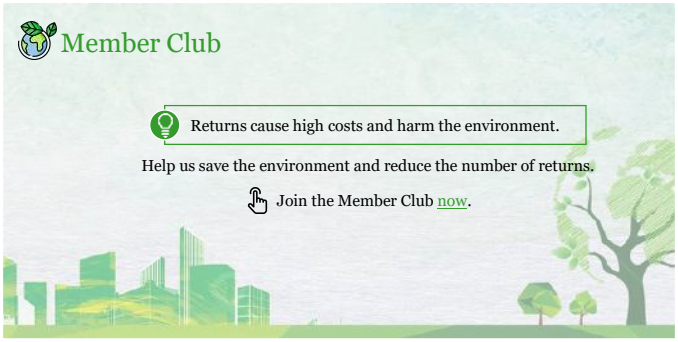


Figure 2. The Displayed Nudge

Following the scenarios, participants were asked about their sociodemographic information, their purchase motivation (PM) and return motivation (RM), as well as their perceived A, C, and R when evaluating the Fashionstore24 Member Club. The respondents then had to indicate their environmental consciousness (EC). Finally, a realism check was conducted to ascertain respondents' perceptions of the authenticity of the scenarios. Responses to all questions were provided on a seven-point Likert scale (ranging from 1="strongly disagree" to 7="strongly agree"). Established scales from previous literature were used to measure all constructs (A, C, R, PM, RM, and EC). For A, C, and R, we used the scales from Xi and Hamari (2019). These questions were typically something like, "I feel free from outside pressures

when I am using the Fashionstore24 Member Club" [A], "I feel like an expert in the Fashionstore24 Member Club" [C], "When I use the Fashionstore24 Member Club, I feel supported by other users" [R]. For PM and RM we used scales from van der Heijden (2003) and van der Heijden et al. (2003), e.g., "I have a positive attitude towards buying products [product returns] on Fashionstore24 Member Club". For EC, we used the Green Scale by Haws et al. (2014), e.g., "I am concerned about wasting the resources of our planet". Prior to the field phase, we pre-tested the questionnaire to ensure the comprehensibility and logic. The final survey took place in November 2023 and resulted in 1000 completed questionnaires. During data cleaning, participants who answered inaccurately to control or attention test questions were excluded from further analysis. The final dataset consists of 988 completed questionnaires. Thanks to the Prolific panel, we were able to obtain a sample that is representative of a large portion of society in terms of age, gender, and education. It is also worth noting that the results of the realism check indicated that the respondents found the presented scenarios to be realistic. On a Likert scale with a neutral midpoint of 4 and a maximum of 7 (i.e., "strongly agree" that the scenario was realistic) the CG averaged 5.65 and the NG averaged 5.19.

Validity and Reliability Assessment

We assessed the validity and reliability of the model by conducting an exploratory factor analysis (EFA) to extract and validate the most important independent factors (i.e., the latent variables). The results of the EFA indicated that each observed variable loaded sufficiently on its underlying latent variable (A, C, R, PM, or RM), with all loadings being significant for both the CG and the NG. We followed recommendations from the EFA literature to ensure an appropriate selection of observed variables (e.g. Conway and Huffcutt 2003). Specifically, we used the maximum likelihood method with oblique rotation (direct oblimin). Oblique rotation was applied because we assume correlations between A, C, R, RM, and PM in our theoretical model. In the course of the EFA "RM I" was excluded due to cross-loadings (< .40) on two latent variables. The Kaiser-Meyer-Olkin measure of sampling adequacy was .945, indicating a "marvelous" factor analysis (Field 2013), and Bartlett's test of sphericity was significant ($p < .001$). We assessed convergent validity through average variance extracted (all values > .50). Composite reliability (all values > .80) and Cronbach's alpha (all values > .80) were calculated to assess internal consistency. To investigate Mod1 and Mod2, we conducted a moderation analysis with the environmental consciousness scores of the survey participants. We did not estimate a latent variable for EC because a moderation analysis with a double mean-centered latent EC variable (as is best practice) resulted in "Heywood cases". Instead, we used the average of the observed variables ([EC I + … + EC VI]/6) in our moderation analysis.

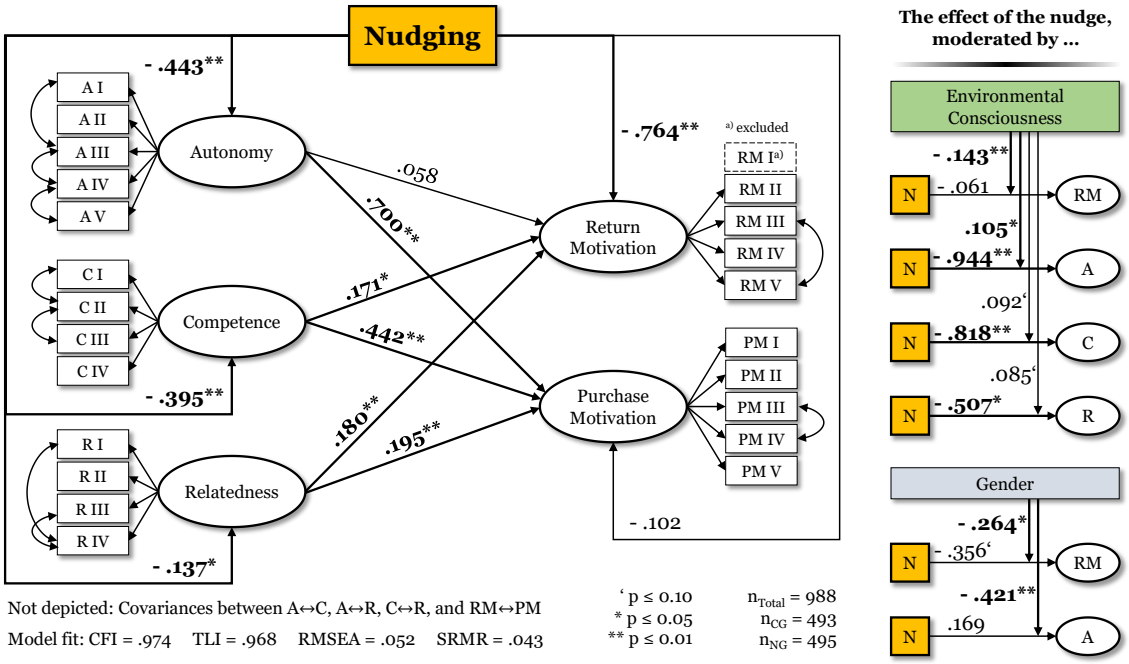


Figure 3. The Fitted Model With Moderation Analysis

Model Fit Statistics

To test our hypothesized research model, we used structural equation modeling (SEM). SEM allows the simultaneous evaluation of both the measurement model and the structural model. We used the R package "lavaan". The assessment of normality using the "mardia" test clearly showed non-normality. Hence, we employed the robust "MLMV" estimator (mean and variance corrections) for our model. According to Gao et al. (2020), this estimator often produces the most accurate results. However, it should be noted that our model also delivers acceptable to good results with other robust or even non-robust estimators. The computed model fit indices indicate that the model fits the observed data well: The comparative fit index (CFI) = .974 and Tucker Lewis index (TLI) = .968 indicate an excellent fit (≥ .95; West et al. 2012). Root mean square error of approximation (RMSEA) values ≤ .05 are good and values between .05 and .08 are acceptable according to Fabrigar et al. (1999). The RMSEA = .052 of our model can therefore be considered nearly good. Diamantopoulos and Siguaw (2000) suggest that values ≤ .05 are acceptable for the standardized root mean square residual (SRMR). Accordingly, our SRMR = .043 has an acceptable fit. It is worth noting that our model still fits quite well, even when we exclude the covariances between the error terms (CFI: 0.959, TLI: 0.951, RMSEA: 0.064, SRMR: 0.049). We can therefore conclude that our SEM (see Figure 3) is probably valid and that our theorized model fits the collected data well.

Results

Figure 4 shows that the nudge reduced the return motivation by about one step from 4.48 to 3.52 on the measured seven-point Likert scale (1=no RM, to 7=very strong RM). This means that the return motivation decreased from a slightly above-neutral stance to a slightly below-neutral stance. Note, that for better readability, all descriptive results in Figure 4 are based on the average of the observed variables (RM II–V in the case of return motivation), while the correlations in Figure 3 are based on the latent variables. Purchase motivation (5.16 to 4.66), perceived autonomy (5.45 to 4.97), and competence (4.98 to 4.53) decreased by about half a step. Participants' perceived relatedness (3.98 to 3.83) and environmental consciousness (EC) (5.03 to 4.83) showed little change. Note, that we asked our questions about the EC at the end of the questionnaire. The nudge seems to have made the participants more aware of their environmental shortcomings (p=.022). However, the observed effect is weak, which helps our analysis because it indicates that a moderation analysis with the environmental consciousness scores is reliable.

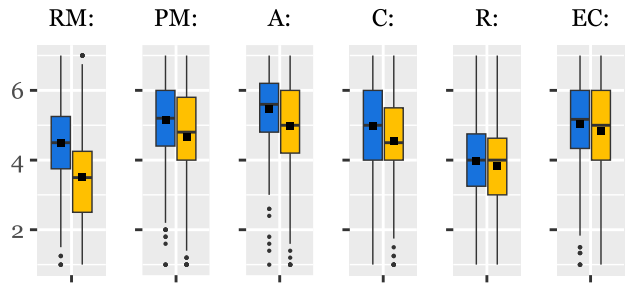


Figure 4. Descriptive Statistics (CG vs. NG)

The fitted structural equation model (Figure 3) estimates that there is a significant direct negative effect of the nudge on RM (-.764), but no significant direct effect on PM (-.102). However, the total effect of the nudge, including indirect effects, is much larger (as seen in Figure 4). In fact, there are several significant indirect mediated effects of the nudge through the three dimensions of the SDT (A: .700, C: .442, and R: .195). That is, the higher the perceived A/C/R, the higher the PM. The effects on RM are weaker (A: .058, C: .171, and R: .180), and perceived autonomy does not seem to influence return motivation at all. Note, that since the nudge has a rather strong negative effect on perceived autonomy (-.443) and competence (-.395), and a small negative effect on perceived relatedness (-.137), the purchase motivation actually decreases significantly (the total effect is: -.102 + -.443*.70 + -.395*.442 + -.137*.195 = -.613). The significance of this nudging effect is completely mediated by the three dimensions A, C, and R. Therefore, the SDT appears to be an appropriate theory to explain the change in PM. The total effect of the nudge on RM (-.882) is larger than the effect on PM (-.613), but not by much. It is interesting that the effect on return motivation is primarily a direct effect, while the effect on purchase motivation is primarily an indirect effect. Thus, the negative effect on RM is likely to be extrinsic in nature, while the negative effect on PM is likely to be intrinsic in nature. Accordingly, we accept our hypotheses H2 (Nudging→A/C/R↓→PM↓) and H3 (Nudging→RM↓) but must reject H4 (Nudging→PM↓). We can only partially accept H1 (Nudging→A/C/R↓→RM↓) because A→RM↑ is not supported.

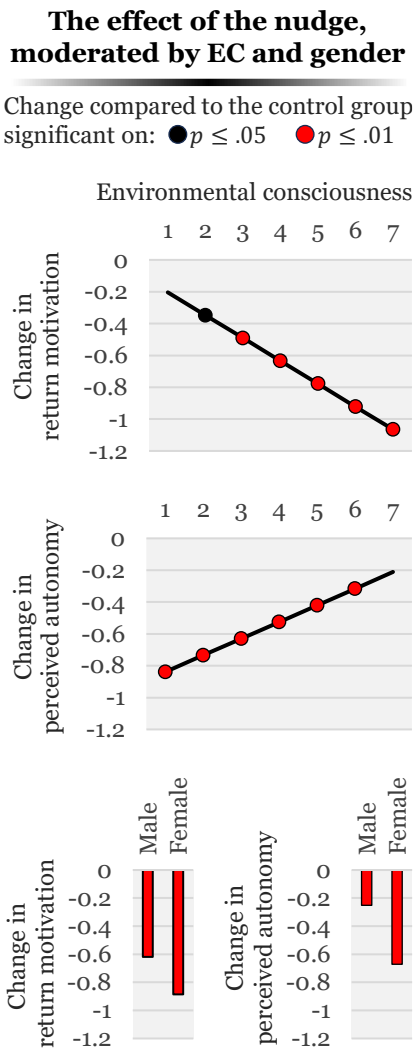


Figure 5. Moderation Analysis

Both Figure 3 and Figure 5 provide information about our moderation analysis. Figure 3 reports the effect sizes of the significant moderations, and Figure 5 visualizes this information graphically. Our moderation tests with demographic characteristics (age, gender, education) only revealed significant interaction effects with the gender of the respondents (1=male, 2=female, non-binary gender excluded). The effect sizes in Figure 3 are based on the SEM model shown in Figure 3 plus EC or gender as a moderator between Nudging→A/C/R/PM/RM. The effect of the nudge on RM is stronger for women than for men (-.264 difference), and the effect on perceived autonomy is also stronger for women (-.421 difference). That is, the intended decreasing effect of nudging on return motivation is stronger for women. However, this is accompanied by a significantly stronger reduction in perceived autonomy, which in turn may be negative, as it may reduce purchase motivation.

Environmental consciousness (measured on a seven-point Likert scale: 1 = low EC, to 7 = high EC) is likely to moderate every observed nudging effect. The effect of the nudge on RM (-.143) is stronger (i.e., more negative) the more environmentally conscious a person is. The effects of the nudge on the perceived autonomy (.105), competence (.092), and relatedness (.085) are less strong (i.e., less negative) the more environmentally conscious a person is. Consequently, both our moderation hypotheses Mod1 and Mod2 are supported. However, the interaction effect is less clear in the case of perceived competence (p=.067) and relatedness (p=.089). It should be noted, that there is no significant correlation between gender and environmental consciousness in our sample. Therefore, the interaction effects of gender and EC outlined above are likely to coexist in parallel.

Discussion

We see that our results are largely in line with our expectations: The less the needs for autonomy, competence, and relatedness are satisfied, the less our participants want to buy from and return items to the fictitious Fashionstore24 Member Club (online shop). Nevertheless, there are differences. Purchase motivation is much more influenced by need satisfaction than by return motivation. Note, that our questions about A, C, and R were rather general. Since the displayed nudge was only about product returns, it is reasonable to assume that most of the variation in perceived A/C/R is due to the new information about product returns. It is interesting that a nudge about product returns changes perceived A/C/R, but that this change in perception is only weakly correlated with RM, but much more strongly correlated with PM. This could be an indication that some survey participants were not necessarily dissatisfied with the content of the nudge, but were generally dissatisfied with being shown a nudge. As a negative side effect, the manipulative nudge may have made the fictitious online shop less attractive overall. In fact, nudging could potentially even backfire if many customers become annoyed and return more items out of spite (Osman et al. 2020). However, this does not appear to have been the case in our study. Instead, it may be that people on average responded in a timid way because they feel bad about their environmentally harmful actions and want to do better. The nudge, with its negative connotation (returns harm the environment), may have made the idea of returning items less appealing to the participants. That is, their intrinsic motivation to return items may have decreased. If people think it is their fault that their A/C/R needs are not being satisfied, they may want to increase their need satisfaction themselves. By returning less, they may hope to become more competent as a person and hope to become a more valuable member of society. Of course, the easier alternative is not to buy from the online shop in the first place. One does not need to return items one did not buy. This reasoning seems to be consistent with the strong decrease in PM seen in our data.

The correlations between Nudging→A/C/R, indicate that nudging has only a weak negative effect on perceived relatedness (-.137), a fairly strong negative effect on perceived competence (-.395), and the strongest negative effect on perceived autonomy (-.443). Intuitively, it can be assumed that the nudge causes individuals who perceive themselves as not eco-friendly to feel less related to other customers of the shop and to society in general. The weakness of the observed effect may be because the survey participants had no prior history with the fictitious online shop. It often takes time to build meaningful relationships. The decrease in perceived competence can be explained on a personal level. Individuals may want their actions to be perceived as competent. Since environmentally harmful behavior is often perceived as incompetent (Schubert 2017), the nudge is likely to reduce perceived personal competence.

The decrease in autonomy is the strongest, which makes sense because nudging as a paternalistic instrument is directly related to perceived autonomy (Thaler and Sunstein 2003). Nudges could be perceived as reducing freedom of choice (Osman et al. 2020) or as psychological pressure (Wachner et al. 2020). Surprisingly, the correlation between perceived autonomy and return motivation is not significant. In contrast, the correlation between perceived autonomy and purchase motivation is the second strongest in our model (.700). This may indicate that dissatisfaction in perceived A, C, and R primarily leads to dissatisfaction with the online shop in general. Based on our data, we conclude that nudging has a significant indirect negative effect on PM (-.511), which is mediated through the three SDT dimensions. The mediated indirect effect of the nudge on RM, however, is much weaker (-.118). Conversely, the nudge has no significant direct effect on PM, but a significant direct effect on RM. Thus, a nudge such as the one we used in our study is likely to act directly as extrinsic motivation on return behavior, but is also likely to decrease the number of purchases indirectly through decreased intrinsic motivation.

This difference could also lead to different long-term effects. The extrinsic effect of nudging on return motivation may be rather short-term, while the lower intrinsic purchase motivation could potentially hurt the shop in the long run (Ryan and Deci 2000). This should be a concern for any online shop thinking about using green nudging. However, the SDT also states that extrinsic motivation can be internalized. In such a case, a lower return rate would no longer be the result of external nudging pressure, but because the individual identifies with the importance of fewer product returns.

Our moderation analysis also shows that some customers are more receptive to the nudge than others. In our sample, environmentally conscious people were more likely to accept a green nudge. Moreover, women seem to be more receptive to the nudge. However, their perceived autonomy was also decreased to a greater extent. Thus, online shops should carefully consider when to use a green nudge. Depending on the type of shop and the type of customer, a digital green nudge may be beneficial or detrimental.

Conclusion, Limitations, and Outlook

Following Osman et al. (2020), we conclude that a green nudge could very well lead to a case where the treatment is offset by negative side effects. This means, that the intended goal of reducing returns is probably accompanied by a reduction in purchases from the online shop. It is unclear whether a green nudge like the one in our study would be financially profitable for online shops. If the nudge primarily affects selection orders, impulsive buying or abusive return policies, then it does not necessarily need to be detrimental to companies if the number of such purchases declines. Moreover, the profitability of green nudges also depends on how many customers will internalize the extrinsic nudging pressure. Our fitted model indicates that the nudge reduces return motivation mainly through extrinsic pressure. Conversely, the negative side effect, i.e., the reduction in purchase motivation, seems to be caused by intrinsic dissatisfaction. This is unfortunate because extrinsic pressure often has only a short-term effect. The question now arises as to how online shops could reduce the probability of such negative side effects. If possible, the green nudge should not always be displayed but should be displayed dynamically, depending on the specific situation (customer, products, contents of the shopping cart, …). Especially, in the context of digital nudging in e-commerce, this seems to be a promising strategy because e-tailers often have extensive information about their customers (e.g., demographics, past purchases and return behavior).

However, our results should also be treated with due caution. Our data is based on a survey-based experiment and not a real-world experiment. Moreover, our survey participants did not have time to develop a meaningful relationship with the fictitious online shop. It is also important to note that our results are about return and purchase motivation and not about actual return and purchase behavior.

Nevertheless, customers' feelings are very important, and our results show that even a simple picture with a green nudge can evoke feelings in digital decision environments and is therefore powerful, but also potentially dangerous.

References

Ambilkar, P., Dohale, V., Gunasekaran, A., and Bilolikar, V. 2022. "Product Returns Management: A Comprehensive Review and Future Research Agenda," *International Journal of Production Research* (60:12), pp. 3920–3944.

ApprissRetail. 2024. "Consumer Returns in the Retail Industry 2020, 2021, 2022, 2023," *Appriss Retail*. (https://apprissretail.com/resources/asset-type/research/, accessed February 25, 2024).

Bear, G. G., Slaughter, J. C., Mantz, L. S., and Farley-Ripple, E. 2017. "Rewards, Praise, and Punitive Consequences: Relations with Intrinsic and Extrinsic Motivation," *Teaching and Teacher Education* (65), pp. 10–20.

Bergram, K., Djokovic, M., Bezençon, V., and Holzer, A. 2022. "The Digital Landscape of Nudging: A Systematic Literature Review of Empirical Research on Digital Nudges," in *CHI Conference on Human Factors in Computing Systems*, pp. 1–16.

Brand, B. M., and Kopplin, C. S. 2023. "Effective Return Prevention Measures in the Post-Purchase Stage: A Best-Worst Scaling Approach," *Marketing ZFP* (45:1), pp. 30–47.

Carrel, C., Gavard-Perret, M.-L., and Caldara, C. 2023. "Factors of Effectiveness of Green Nudges for More Eco-Responsible Behaviour – Systematic Review and Research Directions," *Recherche et Applications En Marketing (English Edition)* (38:3), pp. 32–76.

Chaleshtari, A. S., Elahi, E., Fathi, M., and Khakifirooz, M. 2022. "Impact of Return Leniency on Retailers' Profit and Social Welfare in Monopoly Markets," *Computers & Industrial Engineering* (169).

Clot, S., Della Giusta, M., and Jewell, S. 2022. "Once Good, Always Good? Testing Nudge's Spillovers on Pro Environmental Behavior," *Environment and Behavior* (54:3), pp. 655–669.

Conway, J. M., and Huffcutt, A. I. 2003. "A Review and Evaluation of Exploratory Factor Analysis Practices in Organizational Research," *Organizational Research Methods* (6:2), pp. 147–168.

Cossatin, A. G., Mauro, N., and Ardissono, L. 2024. "Promoting Green Fashion Consumption Through Digital Nudges in Recommender Systems," *IEEE Access* (12), pp. 6812–6829.

Deci, E. L., and Ryan, R. M. 1985. *Intrinsic Motivation and Self-Determination in Human Behavior*, Boston, US: Springer US.

Demarque, C., Charalambides, L., Hilton, D. J., and Waroquier, L. 2015. "Nudging Sustainable Consumption: The Use of Descriptive Norms to Promote a Minority Behavior in a Realistic Online Shopping Environment," *Journal of Environmental Psychology* (43), pp. 166–174.

Diamantopoulos, A., and Siguaw, J. 2000. *Introducing LISREL*, London, UK: SAGE Publications, Ltd.

Fabrigar, L. R., Wegener, D. T., MacCallum, R., and Strahan, E. 1999. "Evaluating the Use of Exploratory Factor Analysis in Psychological Research.," *Psychological Methods* (4:3), pp. 272–299.

Field, A. P. 2013. *Discovering Statistics Using IBM SPSS Statistics: And Sex and Drugs and Rock "n" Roll*, Los Angeles, US: Sage.

Gao, C., Shi, D., and Maydeu-Olivares, A. 2020. "Estimating the Maximum Likelihood Root Mean Square Error of Approximation (RMSEA) with Non-Normal Data: A Monte-Carlo Study," *Structural Equation Modeling: A Multidisciplinary Journal* (27:2), pp. 192–201.

Geng, J., Long, R., Chen, H., and Li, W. 2017. "Exploring the motivation-behavior gap in urban residents' green travel behavior: A theoretical and empirical study," *Resources, Conservation and Recycling*, (125), pp. 282–292.

Ghose, A., Lee, H. A., Nam, K., and Oh, W. 2023. "The Effects of Pressure and Self-Assurance Nudges on Product Purchases and Returns in Online Retailing: Evidence from a Randomized Field Experiment," *Journal of Marketing Research*.

Haws, K. L., Winterich, K. P., and Naylor, R. W. 2014. "Seeing the World through GREEN-tinted Glasses: Green Consumption Values and Responses to Environmentally Friendly Products," *Journal of Consumer Psychology* (24:3), pp. 336–354.

van der Heijden, H. 2003. "Factors Influencing the Usage of Websites: The Case of a Generic Portal in The Netherlands," *Information & Management* (40:6), pp. 541–549.

van der Heijden, H., Verhagen, T., and Creemers, M. 2003. "Understanding Online Purchase Intentions: Contributions from Technology and Trust Perspectives," *European Journal of Information Systems* (12:1), pp. 41–48.

Khan, A. N. 2023. "Elucidating the Effects of Environmental Consciousness and Environmental Attitude on Green Travel Behavior: Moderating Role of Green Self-efficacy," *Sustainable Development*.

Lin, L., Dahl, D. W., and Argo, J. J. 2013. "Do the Crime, Always Do the Time? Insights into Consumer-to-Consumer Punishment Decisions," *Journal of Consumer Research* (40:1), pp. 64–77.

Martínez-López, F. J., Li, Y., Feng, C., Liu, H., and López-López, D. 2022. "Reducing Ecommerce Returns with Return Credits," *Electronic Commerce Research* (23:4), pp. 2011–2033.

Michels, L., Ochmann, J., Günther, S. A., Laumer, S., and Tiefenbeck, V. 2022. "Empowering Consumers to Make Environmentally Sustainable Online Shopping Decisions: A Digital Nudging Approach," in *Proceedings of the 55th Hawaii International Conference on System Sciences 2022*.

Mirbabaie, M., Marx, J., and Erle, L. 2023. "Digital Nudge Stacking and Backfiring: Understanding Sustainable E-Commerce Purchase Decisions," *Pacific Asia Journal of the Association for Information Systems* (15), pp. 65–104.

Nafziger, J. 2020. "Spillover Effects of Nudges," *Economics Letters* (190).

Optoro. 2022. "2022 Impact Report," *Optoro*. (https://www.optoro.com/returns-content/, accessed February 17, 2024).

Osman, M., McLachlan, S., Fenton, N., Neil, M., Löfstedt, R., and Meder, B. 2020. "Learning from Behavioural Changes That Fail," *Trends in Cognitive Sciences* (24:12), pp. 969–980.

Peters, L., and Wieland, D. 2023. "I Tell You What You Want, What You Really Really Want: Digital Information Nudging as a Bridge towards a Sustainable Purchase Decision," in *Proceedings of the Americas Conference on Information Systems 2023*.

Rokonuzzaman, M., Iyer, P., and Harun, A. 2021. "Return Policy, No Joke: An Investigation into the Impact of a Retailer's Return Policy on Consumers' Decision Making," *Journal of Retailing and Consumer Services* (59).

Ryan, R. M., and Deci, E. L. 2000. "Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being.," *American Psychologist* (55:1), pp. 68–78.

Ryan, R. M., Patrick, H., Deci, E. L., and Williams, G. C. 2008. "Facilitating Health Behaviour Change and Its Maintenance: Interventions Based on Self-Determination Theory," *European Health Psychologist* (10:1), pp. 2–5.

Schubert, C. 2017. "Green Nudges: Do They Work? Are They Ethical?," *Ecological Economics* (132), pp. 329–342.

Schürmann, C., Helinski, C., Koch, J., Westmattelmann, D., and Schewe, G. 2023. "Digital Nudging to Promote Sustainable Consumer Behavoir? An Experimental Analysis in Online Fashion Retail," in *Proceedings of the 31st European Conference on Information Systems 2023*.

Thaler, R. H., and Sunstein, C. R. 2003. "Libertarian Paternalism," *American Economic Review* (93:2), pp. 175–179.

Thaler, R. H., and Sunstein, C. R. 2008. *Nudge: Improving Decisions About Health, Wealth and Happiness*, Yale University Press.

Tian, X., and Sarkis, J. 2022. "Emission Burden Concerns for Online Shopping Returns," *Nature Climate Change* (12:1), pp. 2–3.

Wachner, J., Adriaanse, M. A., and De Ridder, D. T. D. 2020. "And How Would That Make You Feel? How People Expect Nudges to Influence Their Sense of Autonomy," *Frontiers in Psychology* (11).

Wang, J., Yang, X., Bailey, A., and Wang, J. 2021. "Positive Spillover of Consumers' Sustainable Behaviors: The Mediating Role of Self-Determination Need Satisfaction," *Journal of Cleaner Production* (317).

Wang, Y., Anderson, J., Joo, S.-J., and Huscroft, J. R. 2019. "The Leniency of Return Policy and Consumers' Repurchase Intention in Online Retailing," *Industrial Management & Data Systems* (120:1), pp. 21–39.

Weinmann, M., Schneider, C., and Brocke, J. V. 2016. "Digital Nudging," *Business & Information Systems Engineering* (58:6), pp. 433–436.

West, R. F., Meserve, R. J., and Stanovich, K. E. 2012. "Cognitive Sophistication Does Not Attenuate the Bias Blind Spot.," *Journal of Personality and Social Psychology* (103:3), pp. 506–519.

Xi, N., and Hamari, J. 2019. "Does Gamification Satisfy Needs? A Study on the Relationship between Gamification Features and Intrinsic Need Satisfaction," *International Journal of Information Managemen*t (46), pp. 210–221.

von Zahn, M., Bauer, K., Mihale-Wilson, C., Jagow, J., Speicher, M., and Hinz, O. 2022. "The Smart Green Nudge: Reducing Product Returns through Enriched Digital Footprints & Causal Machine Learning," *SAFE Working Paper* No. 363.